Agnibh Dasgupta

adg002@gmail.com | Website | Github | Linkedin | +1 4357549599 | Omaha, NE, USA

Education

2021 - 2026 PhD, Computing & Information Science, **University of Nebraska, Omaha**
2018 - 2020 Master of Science, Computer Science, **Utah State University**
2013 - 2017 Bachelor of Technology, Computer Science, **West Bengal University of Technology**

Publications

2026 **TIACam: Text-Anchored Invariant Feature Learning with Auto-Augmentation for Camera-Robust Zero-Watermarking (CVPR)** *(in press)*
- Combined **CLIP image–text features** with a **learned augmentation module**
- Exceeded state-of-the-art watermark robustness to camera and environmental distortions

2025 **Watermarking Language Models through Language Models (IEEE TAI)**
- Built a prompt-based **LLM watermarking** framework without modifying model weights
- Achieved **88%+ watermark detection accuracy** across 25 **open-source LLMs**

2024 **Book Chapter, Imaging Science**
- Improved the scalability of our attention-based watermarking under heavy compound noise

2024 **Leveraging Artificial Intelligence (AI) to Enhance CS Instruction (FIE)**
- Utilized multi-modality in 3D CNN models for mood detection in K-12 classroom videos

2023 **Robust Image Watermarking based on Cross-Attention and ID Learning (CSCI)**
- Designed an attention-based robust image watermarking scheme using **vision transformers**
- Trained a robust invariant domain using **self-supervised learning**

2022 **Perspective transformation layer (CSCI)**
- Proposed a lightweight layer that learns 2D projections of 3D views for multiple viewpoints

Present and Past Research

Present **Invariant Representation Learning in LLMs for Model Attribution** *(under review)*
- Identified semantically meaningful subspace from **9 open-source LLMs** using **PCA**
- Operationalized these representations for **reliable model attribution**

2025 **Utilization of SAM2 Model for the Segmentation of Retinal Vascular Leakage on Ultra-wide-field Fluorescein Angiography in Non-infectious Retinal Vasculitis** (AAO)
- Trained a classifier on patches of large optic scans to accurately detect areas of leakage

2022 **Sentiment Analysis on Twitter data on Electric Vehicles**
- Analyzed sentiment with **lexical and Word2Vec-LSTM** approaches on scraped twitter data

2020 **Deep Q Learning applied to stock trading**
- Built a **reinforcement learning trading system** using auto-scraped S&P index data

Reviewer Service

2022 - Present ACCV, Scientific Reports - Nature, Springer Nature

Teaching Experience

2022 - 2023 **Intern Supervisor**, University of Nebraska at Omaha
2023 - 2025 **Graduate Teaching Assistant**, University of Nebraska at Omaha
Courses: Intro to Python, Capstone
2018 - 2020 **Graduate Teaching Assistant**, Utah State University
Courses: Programming Languages, Intro to C++, Intro to AI

Technical Skills

Programming & Systems:	Python, Bash, Linux, MATLAB, Git, JAX, CUDA
Deep Learning Frameworks:	PyTorch, TensorFlow, Transformers, LLMs, NLP
Distributed & Optimization:	DeepSpeed, PyTorch DDP, tf.distribute, Slurm/HPC